CHAPTER 5

Arrays

Real-world problems usually require the storage, retrieval, and manipulation of related data. If data are related, then it makes sense to keep them collected together. A real data set could be, say, the names of people in a tutorial group. It would be possible to use scalar variables to represent each individual in the group, but this would mean maintaining many separate variables, an overly complex and definitely more error-prone approach. Grouping each value into a set makes more sense, and this is where the `array` data type comes in.

Two fundamental properties of arrays in PHP are that they are *countable* (they have cardinality; the number of elements in the array can be measured) and that they are *iterable* (capable of having an operation repeated on each element). A synonym of iterable is "traversable," and you will often see these terms used when discussing arrays. `Countable`, `Iterable`, and `Traversable` are reserved words in PHP 8: `Countable` and `Traversable` are interfaces for objects that can behave like arrays. `Iterable` is a pseudo-type that matches arrays and array-like objects.

Here we look at various techniques for manipulating arrays; PHP provides a plethora of built-in tools for working with arrays, as we shall see. These are the topics covered in this chapter:

- Indexed arrays
- Associative arrays
- Multidimensional arrays
- Accessing array elements
- ArrayObject
- Adding, removing, and changing array elements
- Iterating arrays
- Extracting data from arrays

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